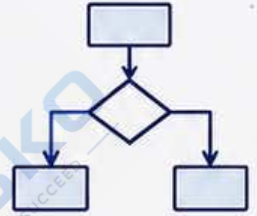


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Question Bank

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Computer Network and Advance Python Programming

Unit I - Introduction to Networking

- **Answers in one or two lines (1 mark questions)**

1. Define Computer Network.
2. What is Computer Networking?
3. What is the need for a Computer Network?
4. Define Communication Protocol.
5. What is LAN?
6. What is MAN?
7. What is WAN?
8. Expand LAN, MAN and WAN.
9. Define Network Topology.
10. Name the different types of Network Topologies in the syllabus.
11. What is Bus Topology?
12. What is Ring Topology?
13. What is Star Topology?
14. What is Mesh Topology?
15. Which topology uses a central connecting device?
16. Which topology uses a backbone cable?
17. Which topology connects nodes in the form of a ring?
18. Which type of network covers a large geographical area?
19. Which type of network is generally used within a building or campus?
20. What is the main purpose of a communication protocol?

● **Short Answer Questions (3 marks questions)**

1. Explain the concept of Computer Networking.
2. Explain the need and importance of Computer Networks.
3. Explain Communication Protocol with suitable examples.
4. Explain LAN with its characteristics and applications.
5. Explain MAN with its characteristics and applications.
6. Explain WAN with its characteristics and applications.
7. Differentiate between LAN, MAN and WAN.
8. Explain Bus Topology with its advantages and disadvantages.
9. Explain Ring Topology with its advantages and disadvantages.
10. Explain Star Topology with its advantages and disadvantages.
11. Explain Mesh Topology with its advantages and disadvantages.
12. Differentiate between Bus and Star Topology.
13. Differentiate between Ring and Mesh Topology.
14. Explain the importance of Network Topology.
15. Explain the applications of Computer Networks.

● **Long Answer Questions (6 marks questions)**

1. Define Computer Network and explain its need and importance in detail.
2. Explain Communication Protocol and its importance in computer networking with suitable examples.
3. Explain LAN, MAN and WAN with their characteristics, applications and suitable diagrams.
4. Differentiate between LAN, MAN and WAN in detail.
5. Explain Bus Topology with a neat diagram, advantages and disadvantages.
6. Explain Ring Topology with a neat diagram, advantages and disadvantages.
7. Explain Star Topology with a neat diagram, advantages and disadvantages.
8. Explain Mesh Topology with a neat diagram, advantages and disadvantages.
9. Explain Ring, Bus, Star and Mesh Topologies with neat diagrams.
10. Compare different Network Topologies based on structure, advantages, disadvantages and applications.

Unit II - Network Devices and OSI Reference Model

- **Answers in one or two lines (1 mark questions)**

1. Define a Repeater.
2. What is a Hub?
3. Define a Switch.
4. What is a Bridge?
5. Define a Router.
6. Expand OSI.
7. What is the OSI Reference Model?
8. How many layers are present in the OSI Model?
9. Name all seven layers of the OSI Model.
10. Which is the lowest layer of the OSI Model?
11. Which is the highest layer of the OSI Model?
12. Which OSI layer is responsible for routing?
13. Which layer provides end-to-end delivery?
14. Which layer is responsible for transmission of raw bits?
15. Which device regenerates or amplifies network signals?
16. Which device forwards frames using MAC addresses?
17. Which device forwards packets using IP addresses?
18. What is the main function of a Hub?
19. What is the main function of a Switch?
20. What is the main function of a Router?

● **Short Answer Questions (3 marks questions)**

1. Explain Repeater and its function.
2. Explain Hub and its working.
3. Explain Switch and its working.
4. Explain Bridge and its function.
5. Explain Router and its function.
6. Differentiate between Hub and Switch.
7. Differentiate between Switch and Router.
8. Differentiate between Bridge and Router.
9. Explain the OSI Reference Model.
10. Explain the Physical Layer and its functions.
11. Explain the Data Link Layer and its functions.
12. Explain the Network Layer and its functions.
13. Explain the Transport Layer and its functions.
14. Explain Session, Presentation and Application Layers.
15. Explain the importance of the OSI Reference Model.

● **Long Answer Questions (6 marks questions)**

1. Explain Repeater, Hub, Switch, Bridge and Router with suitable diagrams and functions.
2. Differentiate between Repeater, Hub, Switch, Bridge and Router.
3. Explain the OSI Reference Model with a neat diagram.
4. Explain all seven layers of the OSI Model and their functions in detail.
5. Explain the Physical, Data Link, Network and Transport layers in detail.
6. Explain the Session, Presentation and Application layers in detail.
7. Explain the working and functions of Hub, Switch, Bridge and Router.
8. Compare Hub, Switch, Bridge and Router based on their functions and operation.
9. Explain the importance and advantages of the OSI Reference Model.
10. Draw a neat diagram of the OSI Model and explain the function of each layer.

Unit III - String Methods, Lists and Dictionaries

- **Answers in one or two lines (1 mark questions)**

1. What is a string in Python?
2. What is string formatting?
3. What is a list?
4. How is a list created in Python?
5. What is list indexing?
6. What is list slicing?
7. Are lists mutable in Python?
8. Name any two built-in functions used with lists.
9. Name any two list methods.
10. What is the `append()` method?
11. What is the `insert()` method?
12. What is the `remove()` method?
13. What is the `pop()` method?
14. What is a dictionary?
15. What is a key-value pair?
16. How is a dictionary created in Python?
17. How can a dictionary value be accessed?
18. How can a dictionary value be modified?
19. What is the `del` statement?
20. Name any two dictionary methods.

- **Short Answer Questions (3 marks questions)**

1. Explain string formatting with a suitable example.
2. Explain how lists are created in Python.
3. Explain basic list operations with examples.
4. Explain list indexing with suitable examples.
5. Explain list slicing with suitable examples.
6. Explain any five built-in functions used on lists.
7. Explain any five list methods with examples.
8. Explain the `append()`, `insert()`, `remove()` and `pop()` methods.
9. Explain dictionaries with a suitable example.
10. Explain how to create and access dictionary key-value pairs.
11. Explain how dictionary values can be modified.
12. Explain built-in functions used on dictionaries.
13. Explain commonly used dictionary methods.
14. Explain the `del` statement with suitable examples.
15. Differentiate between Lists and Dictionaries.

- **Long Answer Questions (6 marks questions)**

1. Explain Lists in Python, including creation, basic operations, indexing, slicing, built-in functions and methods with suitable examples.
2. Explain list indexing and slicing in detail with suitable Python programs.
3. Explain different list methods such as `append()`, `insert()`, `remove()`, `pop()`, `sort()` and `reverse()` with examples.
4. Explain String Formatting in Python with suitable examples.
5. Explain Dictionaries in Python, including creation, accessing and modifying key-value pairs.
6. Explain built-in functions and commonly used dictionary methods with suitable examples.
7. Explain the `del` statement for deleting elements from Lists and Dictionaries.
8. Differentiate between Lists and Dictionaries with suitable examples.
9. Explain Lists and Dictionaries and compare their operations and methods.

10. Write a Python program demonstrating list operations, list methods, dictionary operations and dictionary methods.

Unit IV - Tuples and Sets

● Answers in one or two lines (1 mark questions)

1. What is a Tuple in Python?
2. How do you create a Tuple?
3. Are Tuples mutable or immutable?
4. What is Tuple indexing?
5. What is Tuple slicing?
6. Name any two built-in functions used on Tuples.
7. Name the methods available for Tuples.
8. What is a Set in Python?
9. Can a Set contain duplicate elements?
10. How do you create an empty Set?
11. What is Set traversal?
12. What is a Frozen Set?
13. What is the `zip()` function?
14. Can a Tuple contain elements of different data types?
15. Can a Set be indexed?
16. Name any two Set methods.
17. What is the difference between a Set and a Frozen Set?
18. What is the relation between Tuples and Lists?
19. What is the relation between Tuples and Dictionaries?
20. What does the `len()` function return when applied to a Tuple?

- **Short Answer Questions (3 marks questions)**

1. Explain Tuples in Python with a suitable example.
2. Explain how to create Tuples.
3. Explain basic Tuple operations with examples.
4. Explain Tuple indexing with suitable examples.
5. Explain Tuple slicing with suitable examples.
6. Explain built-in functions used on Tuples.
7. Explain Tuple methods with examples.
8. Explain the relationship between Tuples and Lists.
9. Explain the relationship between Tuples and Dictionaries.
10. Explain the `zip()` function with a suitable example.
11. Explain Sets in Python with a suitable example.
12. Explain commonly used Set methods.
13. Explain Set traversal with an example.
14. Explain Frozen Sets with a suitable example.
15. Differentiate between Tuples and Sets.

- **Long Answer Questions (6 marks questions)**

1. Explain Tuples in Python, including creation, basic operations, indexing, slicing, built-in functions and methods with suitable examples.
2. Explain Tuple indexing and slicing in detail with suitable Python programs.
3. Explain the relationship between Tuples and Lists with suitable examples.
4. Explain the relationship between Tuples and Dictionaries with suitable examples.
5. Explain the `zip()` function in detail with suitable examples.
6. Explain Sets in Python, including creation, operations, methods and traversal.
7. Explain different Set methods with suitable examples.
8. Explain Frozen Sets, their characteristics and uses.
9. Differentiate between Tuples, Lists, Sets and Frozen Sets with suitable examples.
10. Write a Python program demonstrating Tuple operations, `zip()` function, Set operations and Set traversal.

Unit I - Introduction to Software Engineering and Software Development Models

- **Answers in one or two lines (1 mark questions)**

1. Define Software.
2. Define Software Engineering.
3. What is a software product?
4. State any two characteristics of software.
5. Name any two types of software.
6. What is system software?
7. What is application software?
8. What is the need for Software Engineering?
9. Expand SDLC.
10. Define Software Development Life Cycle.
11. Name the phases of SDLC.
12. What is a software process model?
13. Define Waterfall Model.
14. Define Prototype Model.
15. Define Spiral Model.
16. Which model follows a sequential development approach?
17. What is a prototype?
18. What is risk analysis in the Spiral Model?
19. Name any two Software Development Models.
20. What is the purpose of SDLC?

● **Short Answer Questions (3 marks questions)**

1. Define Software Engineering and explain its need.
2. Explain the characteristics of software.
3. Explain different types of software.
4. Explain the importance of Software Engineering.
5. Explain Software Development Life Cycle (SDLC).
6. Explain the phases of SDLC.
7. Explain the concept of Software Process Models.
8. Explain the Waterfall Model with its phases.
9. Explain the Prototype Model.
10. Explain the Spiral Model.
11. Differentiate between Waterfall Model and Prototype Model.
12. Differentiate between Waterfall Model and Spiral Model.
13. Explain the advantages of the Waterfall Model.
14. Explain the advantages and disadvantages of the Prototype Model.
15. Explain the advantages and disadvantages of the Spiral Model.

● **Long Answer Questions (6 marks questions)**

1. Define Software Engineering and explain its characteristics, need, and importance.
2. Explain different types of software with suitable examples.
3. Explain SDLC and all its phases in detail with a suitable diagram.
4. Explain Software Development Process Models and their importance.
5. Explain the Waterfall Model with a neat diagram, advantages, and disadvantages.
6. Explain the Prototype Model with a neat diagram, advantages, and disadvantages.
7. Explain the Spiral Model with a neat diagram, advantages, and disadvantages.
8. Compare Waterfall, Prototype, and Spiral Models.
9. Explain the different phases of SDLC and their activities.
10. Explain the Waterfall, Prototype, and Spiral Models and discuss their suitable applications.

Unit II - Software Requirements and Requirements Engineering

- **Answers in one or two lines (1 mark questions)**

1. Define Software Requirement.
2. What is Requirements Engineering?
3. What is a Functional Requirement?
4. What is a Non-functional Requirement?
5. Define User Requirement.
6. Define System Requirement.
7. What is Requirements Elicitation?
8. What is Requirements Analysis?
9. What is Requirements Validation?
10. What is Requirements Management?
11. What is a Software Requirements Specification (SRS)?
12. What is the purpose of requirements engineering?
13. Give one example of a functional requirement.
14. Give one example of a non-functional requirement.
15. What is a stakeholder?
16. Who is a software user?
17. What is requirements documentation?
18. Why is requirements validation required?
19. What is requirements change?
20. Name the major activities of the requirements engineering process.

● **Short Answer Questions (3 marks questions)**

1. Explain Software Requirements.
2. Explain Functional Requirements with examples.
3. Explain Non-functional Requirements with examples.
4. Differentiate between Functional and Non-functional Requirements.
5. Explain User Requirements.
6. Explain System Requirements.
7. Differentiate between User Requirements and System Requirements.
8. Explain Requirements Engineering and its importance.
9. Explain the Requirements Engineering Process.
10. Explain Requirements Elicitation and Analysis.
11. Explain Requirements Validation.
12. Explain Requirements Management.
13. Explain the role of stakeholders in requirements engineering.
14. Explain the importance of requirements documentation.
15. Explain problems associated with changing software requirements.

● **Long Answer Questions (6 marks questions)**

1. Explain Software Requirements and their different types in detail.
2. Explain Functional and Non-functional Requirements with suitable examples.
3. Differentiate between Functional and Non-functional Requirements with examples.
4. Explain User Requirements and System Requirements in detail.
5. Explain the Requirements Engineering Process with a suitable diagram.
6. Explain Requirements Elicitation and Analysis in detail.
7. Explain Requirements Validation, its objectives, and techniques.
8. Explain Requirements Management and its importance.
9. Explain the complete process of Requirements Engineering from elicitation to management.
10. Explain the role of requirements engineering in successful software development.

Unit III - Risk Management and Design Engineering

- **Answers in one or two lines (1 mark questions)**

1. Define software risk.
2. What is risk management?
3. What is risk identification?
4. What is a risk strategy?
5. What is an RMMM Plan?
6. Expand RMMM.
7. What is risk monitoring?
8. What is risk mitigation?
9. Define Design Engineering.
10. What is software design?
11. What is the design process?
12. What is design quality?
13. Define software architecture.
14. What is data design?
15. What is an architectural pattern?
16. What is User Interface Design?
17. What are Golden Rules of UI Design?
18. What is a system model?
19. What is UML?
20. Expand UML.
21. What is a UML diagram?
22. Name any two UML diagrams.
23. What is a Use Case Diagram?
24. What is a Class Diagram?
25. What is risk assessment?

● **Short Answer Questions (3 marks questions)**

1. Explain software risk.
2. Explain the importance of risk management.
3. Explain different software risks.
4. Explain risk identification.
5. Explain risk strategies.
6. Explain the RMMM Plan.
7. Explain the components of an RMMM Plan.
8. Explain the software design process.
9. Explain design quality.
10. Explain Software Architecture.
11. Explain the importance of software architecture.
12. Explain data design.
13. Explain architectural patterns.
14. Explain User Interface Design.
15. Explain the Golden Rules of User Interface Design.
16. Explain the steps involved in interface design.
17. Explain system models.
18. Explain UML and its importance.
19. Explain any three UML diagrams.
20. Differentiate between software architecture and software design.

● **Long Answer Questions (6 marks questions)**

1. Explain Risk Management and its importance in software engineering.
2. Explain different types of software risks and risk identification techniques.
3. Explain risk strategies and the RMMM Plan in detail.
4. Explain the Risk Mitigation, Monitoring, and Management (RMMM) Plan with a suitable example.
5. Explain the Software Design Process and characteristics of a good design.
6. Explain Design Quality and the factors affecting software design quality.

7. Explain Software Architecture, Data Design, and Architectural Patterns.
8. Explain User Interface Design, its Golden Rules, and interface design steps.
9. Explain System Models and UML Diagrams with suitable examples.
10. Explain the complete design engineering process from software design to user interface design.

Unit IV - Inheritance and Interfaces

● Answers in one or two lines (1 mark questions)

1. Define Software Testing.
2. What is a Testing Strategy?
3. What is the purpose of Software Testing?
4. What is Black Box Testing?
5. What is White Box Testing?
6. What is Validation Testing?
7. What is System Testing?
8. What is Debugging?
9. What is a test case?
10. What is a test plan?
11. What is the difference between testing and debugging?
12. What is conventional software testing?
13. What is the objective of validation testing?
14. What is the objective of system testing?
15. Name any two types of software testing.
16. What is a testing strategy?
17. What is a software error?
18. What is a software defect?
19. What is a failure in software testing?
20. What is the role of debugging in software development?

● **Short Answer Questions (3 marks questions)**

1. Explain Software Testing.
2. Explain the importance of Software Testing.
3. Explain a strategic approach to software testing.
4. Explain the major activities involved in a testing strategy.
5. Explain test strategies for conventional software.
6. Explain Black Box Testing with an example.
7. Explain White Box Testing with an example.
8. Differentiate between Black Box Testing and White Box Testing.
9. Explain Validation Testing.
10. Explain the objectives of Validation Testing.
11. Explain System Testing.
12. Explain the objectives of System Testing.
13. Differentiate between Validation Testing and System Testing.
14. Explain Debugging.
15. Explain the difference between Software Testing and Debugging.
16. Explain the characteristics of an effective testing strategy.
17. Explain the role of test cases in software testing.
18. Explain the importance of systematic testing.
19. Explain the process of identifying errors during testing.
20. Explain the relationship between testing and debugging.

● **Long Answer Questions (6 marks questions)**

1. Explain Software Testing and the strategic approach to software testing in detail.
2. Explain the complete testing strategy for conventional software.
3. Explain Black Box Testing and White Box Testing with suitable examples.
4. Differentiate between Black Box Testing and White Box Testing in detail.
5. Explain Validation Testing, its objectives, and process.
6. Explain System Testing with suitable examples.
7. Differentiate between Validation Testing and System Testing.

8. Explain the Art of Debugging and the debugging process in detail.
9. Explain the relationship between Software Testing and Debugging.
10. Explain the complete software testing process from test planning to debugging.